

Leonard Ndwiga

[lndwiga | LinkedIn](#)

[lndwiga | github](#)



Glasgow, United Kingdom

Email: leonmarigi@gmail.com

Mobile: +447425237540

EDUCATION

University of Nairobi

Master of Science in Bioinformatics

Nairobi, Kenya

December 2018

University of Nairobi

Bachelor of Science in Biochemistry

Nairobi, Kenya

December 2014

Howard University

Certificate: Project Management

United States of America

August 2025

University of Washington

Certificate: Project Management in Global Health

United States of America

September 2023

SCIENTIFIC PROFILE

Computational biologist with over eight years of experience applying multi-omics approaches to investigate host-microbe interactions and infectious diseases using high-throughput sequencing data from clinical samples. My work bridges experimental genomics and computational biology to generate biologically meaningful insights from complex clinical sequencing datasets, with expertise spanning genomics, transcriptomics, metatranscriptomics and pathogen genomics. I develop reproducible computational workflows, integrate diverse omics datasets, and translate sequencing data into biological discoveries that advance our understanding of disease mechanisms, biomarker discovery and precision medicine. I thrive in collaborative, multidisciplinary research environments where computational innovation drives biological and clinical discovery.

SELECTED ACHIEVEMENTS

Multi-omics & Computational Biology

- Led the development of computational workflows integrating genomics, transcriptomics and metatranscriptomics to investigate host-microbe interactions using complex clinical sequencing datasets.

Computational Method Development

- Developed reproducible and scalable bioinformatics pipelines using Python, R, Bash, Snakemake and high-performance computing (HPC) environments for genomics, transcriptomics, metagenomics and pathogen surveillance.

Scientific Leadership & Public Health Impact

- Directed sequencing and bioinformatics operations supporting Kenya's national SARS-CoV-2 genomic surveillance programme, leading multidisciplinary teams and delivering genomic evidence to inform public health decision-making.

Research Funding & International Capacity Building

- Secured competitive research funding as Principal Investigator (NIHR/RSTMH) and continues to deliver genomics and bioinformatics training to researchers and public health scientists across more than 15 African countries through Wellcome Connecting Science, Africa CDC and international partners.

Scientific Communication & Research Output

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- Authored peer-reviewed publications in pathogen genomics, transcriptomics, malaria genomics and infectious disease surveillance, contributing to multidisciplinary collaborations across academia, healthcare and industry.

TECHNICAL SKILLS

Computational Biology & Omics Analysis

- Transcriptomic and metatranscriptomic analyses
- Pathogen genomics and microbial genomics
- Differential gene expression and pathway enrichment analyses (GO, KEGG, GSEA)
- Taxonomic profiling and microbial community analysis
- Comparative genomics, molecular epidemiology and phylogenetic analysis
- Multi-omics data integration
- Biomarker discovery and statistical analysis

Bioinformatics & Scientific Computing

- Python, R and Bash programming
- Linux/Unix and High-Performance Computing (HPC)
- Snakemake workflow development
- Workflow automation and optimization
- Git version control

Next-Generation Sequencing & Genomic Analysis

- Illumina and Oxford Nanopore sequencing platforms
- Whole genome, amplicon and RNA sequencing
- Genome assembly, variant calling and phylogenetic analysis
- Sequencing quality control and performance assessment

Experimental Genomics & Molecular Biology

- DNA and RNA extraction from clinical and research specimens
- Nucleic acid quantification and quality assessment (Qubit, NanoDrop, TapeStation/Bioanalyzer)
- PCR, RT-PCR and qPCR assay preparation
- Illumina and Oxford Nanopore library preparation
- Sequencing workflow optimization and laboratory quality assurance

Scientific Leadership & Collaboration

- Experimental design and computational strategy
- Scientific writing and manuscript preparation
- Grant writing and technical reporting
- Cross-disciplinary collaboration
- Training, mentorship and capacity building
- Scientific presentations and stakeholder engagement

WORK EXPERIENCE

Researcher | University of Glasgow – Centre for Virus Research – 27th February 2024 – Present

- Lead the computational analysis of bulk transcriptomic and metatranscriptomic datasets generated from clinical respiratory samples, investigating host

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transcriptional responses and microbial signatures associated with respiratory disease.

- Develop and optimize reproducible computational workflows for transcriptomic, metatranscriptomic and pathogen genomic analyses using Python, R, Bash, Snakemake and HPC environments.
- Perform end-to-end bioinformatics analyses of Illumina and Oxford Nanopore sequencing datasets, including quality assessment, differential gene expression, taxonomic profiling, pathway enrichment and biological interpretation.
- Optimize analytical strategies for low-input, partially degraded RNA obtained from clinical respiratory samples, improving transcript detection and downstream biological analyses.
- Collaborate with academic, NHS and industry partners to communicate findings through scientific presentations, reports and multidisciplinary research projects.
- Produce technical documentation, reports and reproducible analytical records to ensure data integrity and support transparent, reproducible research.
- Present analytical findings to multidisciplinary teams and contribute to project planning, scientific reporting and decision-making.

Bioinformatics Consultant / Trainer - Wellcome Connecting Science (UK), Africa CDC, ASLM & Partners | 2023 – Present

- Lead the development and delivery of advanced genomics and bioinformatics training programmes for researchers, clinicians and public health scientists across more than 15 African countries.
- Develop reproducible computational workflows for pathogen genomics, metagenomics and transcriptomics using Illumina and Oxford Nanopore sequencing platforms.
- Provide technical leadership and scientific mentorship in Linux, high-performance computing (HPC), workflow development and next-generation sequencing (NGS) data analysis, promoting reproducible computational research practices.
- Collaborate with international partners to strengthen bioinformatics capacity, sequencing infrastructure and genomic surveillance capabilities within National Public Health Laboratories.
- Support the implementation of genomics-based surveillance programmes by advising multidisciplinary teams on analytical strategies, workflow optimization and interpretation of sequencing data.

Head of Sequencing Facility | KEMRI-Wellcome Trust Research Programme – 1st

October 2021 – 16th February 2024

- Led the sequencing facility and directed genomics and bioinformatics operations supporting national pathogen genomic surveillance programmes, generating genomic evidence to inform public health policy and outbreak response.
- Managed multidisciplinary teams across laboratory genomics, bioinformatics and public health, overseeing project delivery, workflow implementation and quality assurance for high-throughput sequencing programmes.
- Designed, developed and implemented reproducible computational pipelines for viral, bacterial and parasitic genomics using Python, R, Bash and high-performance computing (HPC) environments.

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- Led end-to-end next-generation sequencing workflows, including experimental design, DNA/RNA extraction, library preparation, Illumina and Oxford Nanopore sequencing, quality assurance and downstream bioinformatics analyses.
- Produced genomic surveillance reports and scientific analyses to support evidence-based public health decision-making and national disease surveillance programmes.
- Developed laboratory SOPs, analytical documentation and standardized sequencing workflows to improve reproducibility, operational efficiency and data governance.
- Provided scientific leadership through mentoring researchers, delivering regional genomics and bioinformatics training, and collaborating with international partners to strengthen genomic surveillance capacity across Africa.

Research Fellow | KEMRI-Wellcome Trust Research Programme - 8th October 2018 – 30th September 2021

- Coordinated genomics components of multi-site malaria research programmes, integrating laboratory workflows with bioinformatics analyses to support drug resistance surveillance.
- Applied statistical genomics and bioinformatics approaches to analyse complex biological datasets and inform clinical and public health research.
- Collaborated with academic and industry partners to evaluate molecular diagnostics and optimise analytical workflows.
- Presented scientific findings to national stakeholders, contributing evidence to support research priorities and public health policy.
- Coordinated sample management, laboratory workflows and data integration across multidisciplinary research teams.
- Developed SOPs, supported project planning and budgeting, and contributed to the successful delivery of multiple concurrent research projects.

RESEARCH GRANTS

Royal Society of Tropical Medicine and Hygiene (RSTMH) | NIHR small Grant | £5000

Project title: Genotyping Novel Plasmodium falciparum drug resistance markers in samples obtained from drug efficacy studies in Kenya

Start: 1st September 2019

End: 31st May 2021

Initiative to Develop African Research Leaders (IDeAL) | £4100

Project title: Molecular Analysis of Antimalarial Resistance Markers in Parasite Samples Obtained From Children Recruited Into a Drug Efficacy Trial in Kwale, Kenya, 2013

Start: 1st September 2016

End: 31st August 2017

LIST OF PUBLICATIONS

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1. Epidemiology of COVID-19 infections on routine polymerase chain reaction (PCR) and serology testing in Coastal Kenya James Nyagwange*, **Ndwiga L.*** et. al. Feb 2022 - **equal contribution**
2. A review of the frequencies of Plasmodium falciparum Kelch 13 artemisinin resistance mutations in Africa **Ndwiga L. et. al.**, 2021
3. The Plasmodium falciparum Rh5 invasion protein complex reveals an excess of rare variant mutations. **Ndwiga L. et. al.**, 2021
4. Targeted Amplicon deep sequencing of ama1 and mdr1 to track within-host *P. falciparum* diversity throughout treatment in a clinical drug trial Kevin Wamae* **Leonard Ndwiga*** et. al. 2022 -**equal contribution**

Full list of publications available at:

<https://pubmed.ncbi.nlm.nih.gov/?term=Leonard+Ndwiga>

SELECTED CONFERENCES & TRAININGS

Trainer, Wellcome Connecting Science (Côte d'Ivoire., September 2026):

Scheduled to deliver genomics and clinical microbiology training to African research and public health scientists

Trainer, Wellcome Connecting Science (The Gambia, South Africa, 2024–2025):

Delivered genomics and clinical microbiology training to African research and public health scientists

Trainer, Pan-African Bioinformatics Programs (Africa CDC / CZI, 2023–2025):

Delivered multi-country training in genomics and outbreak response workflows

Lead Trainer, Genome Analysis (Virtual Program, 8 months| Kenya Institute of Primate Research (KIPRE)): Designed and delivered structured training for researchers analysing whole genome sequencing data